Multilevel Fair Allocation: on fair allocation in hierarchical structures

Maxime LUCET, Nawal Benabbou, Aurélie Beynier, Nicolas Maudet

LIP6, CNRS, Sorbonne Université

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Fair Allocation : motivations

Fair allocation deals with the design of algorithms to allocate fairly some resources. [Ste49; Fol67]

Traditionally, it focused on fairness between individuals, e.g. divorce settlements or inheritance. [LMMS04; CKM+19]

More recently, fair allocation between groups of individuals drew interest, e.g. allocating office spaces to teams of a laboratory. [CFSV19; BCEZ19; VZ23]

Classical Fair Allocation

Fair Allocation: model

A simple model

- a set of agents $\mathcal{N} = \{1, \ldots, n\}$
- a set of indivisible resources $\mathcal{G} = \{g_1, \ldots, g_m\}$
- an agent $i \in \mathcal{N}$ has an additive utility function over $\mathcal{G}$, s.t. $v_i(S) = \sum_{g \in S} v_i(\{g\})$ for $S \subseteq \mathcal{G}$

Allocation

$A : \mathcal{N} \to 2^{\mathcal{G}}$ is an allocation if

- $\cup_{i \in \mathcal{N}} A(i) \subseteq \mathcal{G}$
- $A(i) \cap A(j) = \emptyset$ for any $i, j \in \mathcal{N}$

Fair Allocation: fairness

Researchers have proposed different approaches to fairness :

- Optimization-based fairness
  - maxmin : maximizing the lowest utility
  - leximin : maximizing the lowest utility, subject to that, maximizing the second lowest utility, and so on.
- Axiomatic fairness

Fair Allocation: fairness

Envy-freeness

A popular notion of fairness is envy-freeness.

Envy-freeness (EF)

Allocation A is envy-free if for any pair of agents $i, j \in \mathcal{N}$, we have

$$v_i(A(i)) \geq v_i(A(j))$$

$\Rightarrow$ an EF allocation is not guaranteed to exist : suppose 2 agents, and 1 item.

Envy-freeness up-to-one good (EF1)

Allocation A is envy-free up-to-one good if for any pair of agents $i, j \in \mathcal{N}$, we have

$$v_i(A(i)) \geq v_i(A(j) \setminus \{g\})$$

for some $g \in A(j)$.

Fair Allocation: round robin algorithm

The simplest algorithm to compute an EF1 allocation is the Round Robin picking sequence.

Round Robin algorithm

- Fix a sequence $1, 2, \ldots, n, 1, 2 \ldots$ of length m.
- Iterate over this sequence, letting each agent pick her favorite item among the remaining ones.

Fair Allocation: additional desirable properties

In addition to EF1, one may wish for efficiency, e.g., *Pareto-optimality*.

Pareto-optimality

An allocation A is Pareto-optimal if there exists no allocation A' s.t. $v_i(A'(i)) \geq v_i(A(i))$, and the inequality is strict for at least one agent $i \in \mathcal{N}$.

Is an EF1 + PO allocation guaranteed to exist?

For additive utilities, yes!

[CKM+19] show that the allocation maximizing the Nash welfare is EF1 + PO. Formally,

$$A \in \arg\max_{A'} \Pi_{i \in \mathcal{N}} v_i(A'(i))$$

The existence of a polytime algorithm remains open.[1]

[1] [BKV18] provide a pseudo-polynomial time algorithm to compute an EF1 + PO allocation.

Multilevel Fair Allocation

Multilevel Fair Allocation : motivations

Classical fair allocation is great, but in real-life, agents may belong to groups, which themselves belong to higher-level groups, ultimately forming a hierarchical structure.

- Allocation of resources across a territory
- Allocation of materials in a hierarchical organization, e.g. a university
- Allocation of tasks to robots with organizational roles [GGLP26]

Multilevel Fair Allocation under Additive Preferences

Maxime Lucet Nawal Benabbou Aurélie Beynier Nicolas Maudet

9th *International Conference on Algorithmic Decision Theory (ADT 2026)*

Paper : [LBBM26]

Multilevel Fair Allocation: motivations

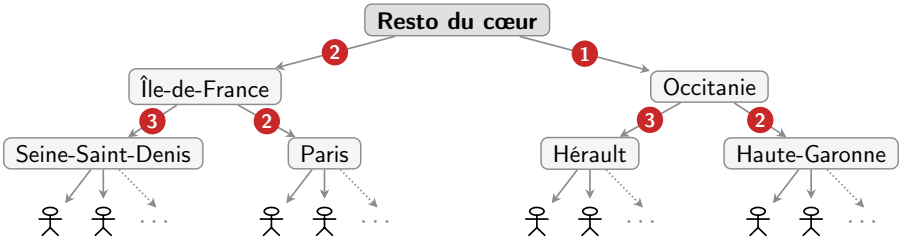


Figure – Simplified structure of Les Restos du coeur.

Multilevel Fair Allocation

We introduce a fair division model within a hierarchical structure, called **multilevel fair allocation**.

Multilevel Fair Allocation : Model

Notations

- a set of entities $\mathcal{N} = \{1, \ldots, n\}$, structured as a tree;
- each node $i \in \mathcal{N}$ has a weight w_i;
- a set of items $\mathcal{G} = \{g_1, \ldots, g_m\}$ to allocate.

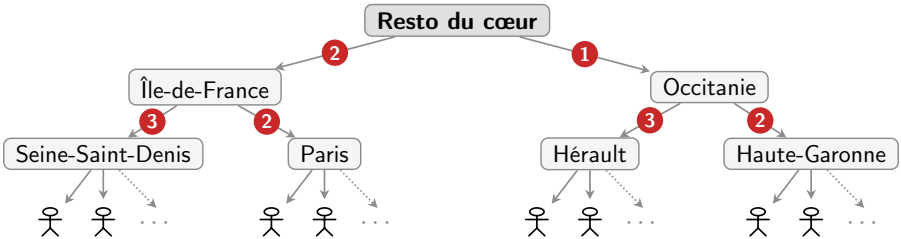


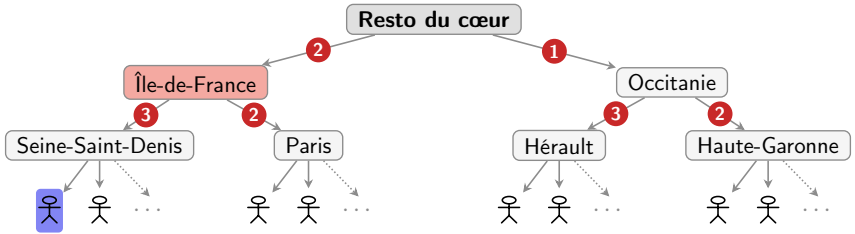
Figure – Simplified structure of Les Restos du coeur.

Definition

A **multilevel allocation** $\pi : \mathcal{N} \to 2^{\mathcal{G}}$ maps any node to a bundle of items such that :

- an internal node receives items only to allocate it to its children.
- given two siblings i, j, we have $\pi(i) \cap \pi(j) = \emptyset$.

We denote $\pi|_{\text{children}(i)}$ the local allocation of i's bundle to its children.



Internal node's utility

The utility v_i of an internal node i is the sum of the utility of its children, i.e.

$$v_i(\pi) = \sum_{j \in \text{children}(i)} v_j(\pi)$$

Leaf's utility

The utility u_x of a leaf x is additive "over the items", i.e.

$$u_x(S) = \sum_{g \in S} u_x(g)$$

Multilevel Fairness

A multilevel allocation π is multilevel fair if, for any internal node i, the allocation $\pi|_{\text{children}(i)}$ is fair.

WEF : weighted envy-freeness

For any pair of agents i, j,

$$\frac{i\text{'s utility}}{i\text{'s weight}} \geq \frac{i\text{'s utility for } j\text{'s bundle}}{j\text{'s weight}}$$

WEF : weighted envy-freeness

For any pair of agents i, j,

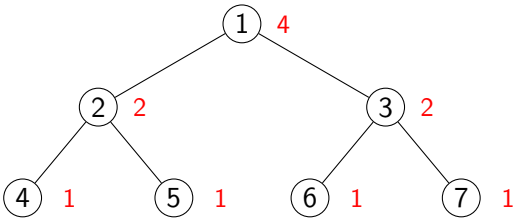
$$\frac{i\text{'s utility}}{i\text{'s weight}} \geq \frac{i\text{'s utility for } j\text{'s bundle}}{j\text{'s weight}}$$

$\Rightarrow$ **Problem** : how does an internal node evaluate the bundle of one of its siblings' bundle?

We assume internal nodes will estimate the value it could derive from a sibling's bundle :

- Optimistic utility $\hat{v}$: computes a *utilitarian-optimal* allocation to its leaves, and measures the utilitarian welfare.
- Pessimistic utility $\overset{\vee}{v}$: computes a *utilitarian-pessimal* allocation to its leaves, and measures the utilitarian welfare.
- Agnostic utility $\overline{v}$: computes the weighted average utility of the bundle to its leaves.

Example



Objets	g_1	g_2	g_3	g_4	g_5
4	2	2	2	2	2
5	1	1	1	1	1
6	2	2	2	2	2
7	1	1	1	1	1

Table – Leaves' preferences

Example of allocation :

$$\left.\begin{array}{c}\pi(4)=\{g_1,g_5\}\\ \pi(5)=\{g_3\}\end{array}\right\}\pi(2)=\{g_1,g_3,g_5\}$$

$$\left.\begin{array}{c}\pi(6)=\{g_2\}\\ \pi(7)=\{g_4\}\end{array}\right\}\pi(3)=\{g_2,g_4\}$$

Estimated utilities

Pessimistic : $\overset{\vee}{v}_3(\{g_1,g_3,g_5\})=3$

Agnostic : $\overline{v}_3(\{g_1,g_3,g_5\})=4.5$

Optimistic : $\hat{v}_3(\{g_1,g_3,g_5\})=6$

Definition

Multilevel allocation π is *estimated multilevel weighted envy-free up-to-one good* for $\mathcal{E} \in \{\text{pess}, \text{agno}, \text{opt}\}$ (denoted by M[$\mathcal{E}$]-WEF1), if for any internal node i, and any pair of its children j, k, we have

$$\frac{v_j(\pi)}{w_j} \geq \frac{\tilde{v}_j(\pi(k) \setminus \{g\})}{w_k}$$

for some $g \in \pi(k)$, where $\tilde{v}$ denotes $\overset{\vee}{v}$, $\overline{v}$ or $\hat{v}$ for $\mathcal{E} = \text{pess}$, agno, or opt, respectively.

Multilevel Fair Allocation: Theoretical and experimental results

Multilevel Fair Allocation: Multilevel Weighted Round Robin

Multilevel extension of the Weighted Round Robin [CISZ21].

Algorithm

At every iteration :

- starting at the root, pick recursively a node that is the least picked child relative to its weight, until a leaf was selected.
- Let this leaf pick her favorite remaining item.
- Repeat until all items are allocated.

Multilevel Fair Allocation: MWRR guarantees M[pess]-WEF1

Theorem

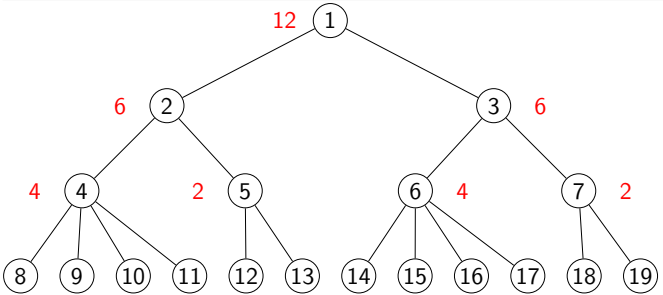
MWRR always returns an M[pess]-WEF1 allocation.

Remark : we show that under identical valuations all three notions coincide, and hence MWRR guarantees our three notions.

Multilevel Fair Allocation: M[agno]-WEF1 is not guaranteed to exist

Theorem

Under general additive valuations, an M[agno]-WEF1 allocation needs not exist.



Item	g_1	g_2	g_3	g_4	g_5
C(4)	2	2	2	2	2
C(5)	1	1	1	1	1
C(6)	2	2	2	2	2
C(7)	1	1	1	1	1

Table – Leaves' preferences over items

Multilevel Fair Allocation: MWRR does not guarantee M[agno]-WEF1

Theorem

Under general additive valuations, MWRR does not guarantee an M[agno]-WEF1 allocation, even when it exists.

$\Rightarrow$ It cannot even be approximated!

Corollary

Under general additive valuations, MWRR does not guarantee an M[opt]-WEF1 allocation, even when it exists.

Multilevel Fair Allocation: Beyond worst-case with experimental tests

Experimental protocol :

- three types of trees : balanced, comb, partially unbalanced.
- either random weights, or dependent on the number of leaves.
- preferences generated with different techniques : mallows, resampling-dirichlet, cost utilities, correlated preferences.

Results

- Out of 200,000 generated instances, only 35 allocations returned by MWRR were not M[agno]-WEF1.
- M[opt]-WEF1 is much more challenging : in some settings, close to 100% allocations returned were not fair.

Multilevel Fair Allocation: take-away message

Multilevel Fair Allocation: take-away message

In this paper, we studied a fair allocation problem within a hierarchical structure. We showed that :

- Adapting fairness notions to the multilevel setting is **not neutral**.
- A multilevel extension of the WRR could guarantee the weakest fairness notion, but not the stronger ones – but behaved very well experimentally.

Open questions and future directions

- Is M[agno]-WEF1 guaranteed to exist under normalized utilities?
- What if the hierarchical organization is a DAG instead of a tree?

Multilevel Fair Allocation with Two-Sided Preferences

Two-sided preferences: motivations

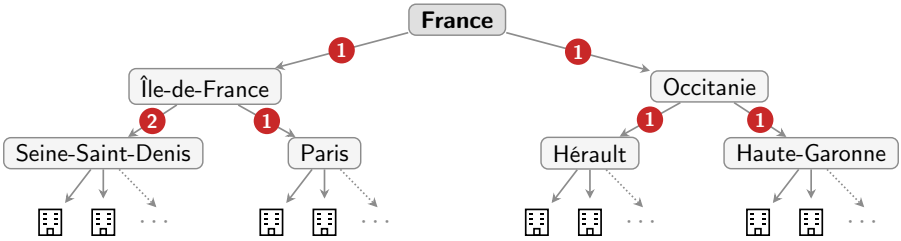


Figure: Territorial distributions of hospitals.

References

This paper is a cross-over between the paper I just presented [LBBM26] and a paper on fairness in many-to-one matchings [IKSS23].

Two-sided preferences: Model

Two-sided preferences: Model

Part of the model is identical to the previous paper:

- a set of entities $\mathcal{N}$, structured in a tree
- each node $i \in \mathcal{N}$ has a weight

Novelty

We do not allocate simple resources, now we allocate individuals with ordinal preferences over the leaves of the tree.

M[pess]-WEF1

Multilevel allocation π is *M[pess]-WEF1*, if for any internal node i, and any pair of its children j, k, we have

$$\frac{v_j(\pi)}{w_j} \geq \frac{\overset{\vee}{v}_j(\pi(k) \setminus \{g\})}{w_k}$$

for some $g \in \pi(k)$.

Weighted balancedness

Multilevel allocation π is *weighted balanced* if for any internal node i and any child j of i, we have

$$\left\lfloor m \cdot \frac{w_j}{w_{\text{children}(i)}} \right\rfloor \leq |\pi(j)| \leq \left\lceil m \cdot \frac{w_j}{w_{\text{children}(i)}} \right\rceil$$

M[pess]-WEF1

Multilevel allocation π is *M[pess]-WEF1*, if for any internal node i, and any pair of its children j, k, we have

$$\frac{v_j(\pi)}{w_j} \geq \frac{\overset{\vee}{v}_j(\pi(k) \setminus \{g\})}{w_k}$$

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Swap stability

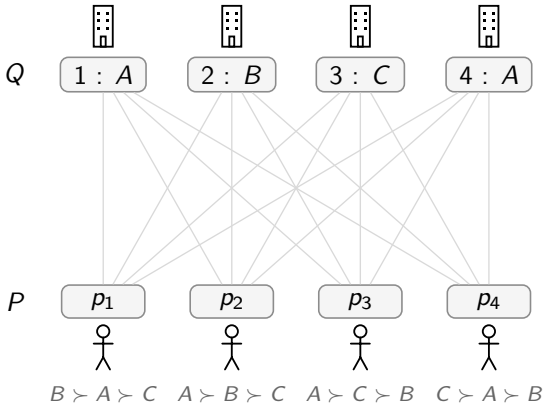
Multilevel allocation π is *swap stable* if there exists no swap of leaves between two individuals s.t. every individuals and leaves involved are not worse-off, and at least one is strictly better-off.

Individual stability

Multilevel allocation π is *individually stable* if no individual can profitable deviate to another leaf, while making neither involved leaves worse-off.

Two-sided preferences: Algorithm

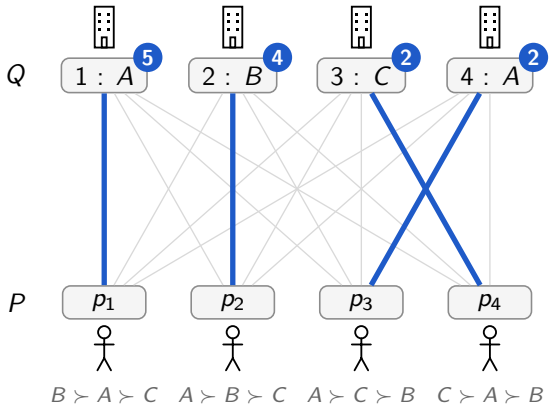
Two-sided preferences: Algorithm



u	p_1	p_2	p_3	p_4
A	5	5	2	1
B	4	4	3	1
C	2	2	2	2

1. Complete bipartite graph G: turns Q (MWRR order) vs. individuals P.

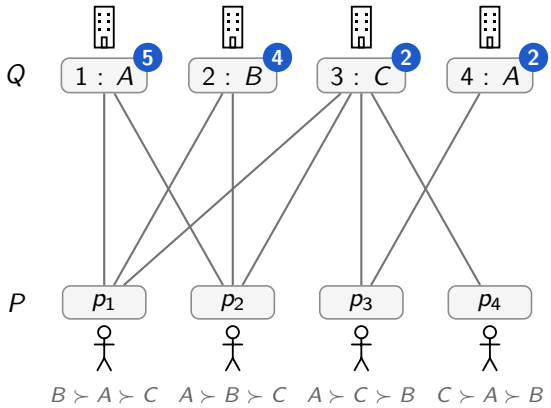
Two-sided preferences: Algorithm



u	p_1	p_2	p_3	p_4
A	5	5	2	1
B	4	4	3	1
C	2	2	2	2

2. μ: lexicographically maximal matching; badge = utility of each turn.

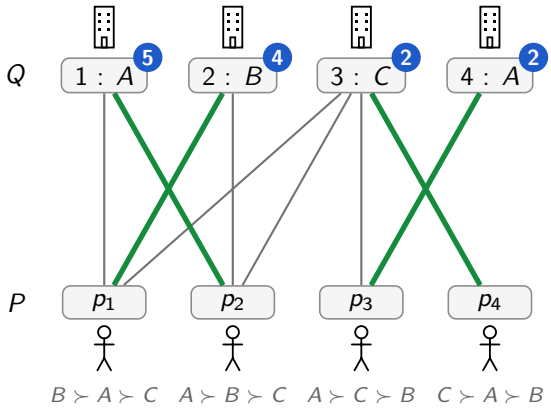
Two-sided preferences: Algorithm



u	p_1	p_2	p_3	p_4
A	5	5	2	1
B	4	4	3	1
C	2	2	2	2

3. G^*: keep only the edges whose weight equals the turn's badge.

Two-sided preferences: Algorithm



u	p_1	p_2	p_3	p_4
A	5	5	2	1
B	4	4	3	1
C	2	2	2	2

4. μ^*: same utilities, sum of ranks 4 instead of 6 for μ.

Two-sided preferences: Algorithm 1

Theorem 1

Algorithm 1 is M[pess]-WEF1, weighted balanced, and swap stable for non-negative individuals.

Theorem 2

Algorithm 1 is M[pess]-WEF1, swap stable, and individually stable for non-negative individuals when adding enough dummy individuals.

Two-sided preferences: Algorithm 1

Theorem 1

Algorithm 1 is M[pess]-WEF1, weighted balanced, and swap stable for non-negative individuals.

Theorem 2

Algorithm 1 is M[pess]-WEF1, swap stable, and individually stable for non-negative individuals when adding enough dummy individuals.

Two-sided preferences: Perspectives

Two-sided preferences: Perspectives

Additional results we have:

- We adapt Algorithm 1 to work with non-positive individuals
- Under binary additive utilities, we obtain M[pess]-WEF1 along with Pareto-optimality with another algorithm

Future directions

- Extend algorithms from [IKK+25] which achieve better properties for the individuals.
- Run some experiments to see if we obtain M[agno]-WEF1 experimentally.

Conclusion

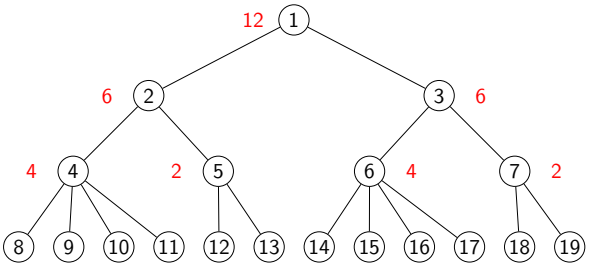
Conclusion

What we covered in this presentation:

- Introduced Fair Division
- Presented *Multilevel Fair Allocation under Additive Preferences*, showing the challenges coming with the multilevel setting, in particular related to the adaptation of existing fairness notions.
- Presented on-going work mixing multilevel fair division and many-to-on matchings: it is possible to achieve great properties for very common allocation problems.

Thanks for your attention

MWRR does not guarantee M[agno]-WEF1



Objets	g_1	g_2	g_3	g_4	g_5
C(4)	5/21	5/21	1/21	5/21	5/21
C(5)	1/5	1/5	1/5	1/5	1/5
C(6)	5/21	5/21	5/21	1/21	5/21
C(7)	1/5	1/5	1/5	1/5	1/5

Table – Leaves' preferences

Returned allocation :

- $\pi(8) = \{g_1\}$
- $\pi(9) = \{g_5\}$
- $\pi(12) = \{g_3\}$
- $\pi(14) = \{g_2\}$
- $\pi(18) = \{g_4\}$

Non-M[agno]-WEF1 allocation

$$v_3(\pi) = 0.438$$

$$\bar{v}_3(\pi(2) \setminus \{g_1\}) = 0.451$$

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